

Coffee Shop Sales Dashboard Project Report

Project Title

Coffee Shop Sales Analysis & Business Intelligence Dashboard

Executive Summary

This project focuses on analyzing transactional sales data from a coffee shop business to identify key business insights, customer purchasing behavior, sales trends, operational performance, and product performance. The project was developed using Excel-based analytics techniques and dashboard reporting methods commonly used in modern business intelligence environments.

The objective of the project was to transform raw sales transaction data into meaningful business insights that support strategic decision-making. The analysis includes revenue trends, store performance, top-selling products, peak sales hours, category-level analysis, and customer purchasing patterns.

The dashboard provides a professional and interactive business overview suitable for stakeholders, managers, and decision-makers.

Project Objectives

The main objectives of this project were:

- Analyze coffee shop sales performance
 - Identify top-performing products
 - Understand customer purchasing patterns
 - Measure store-level performance
 - Detect peak operational hours
 - Track monthly sales growth
 - Build KPI-driven business dashboards
 - Improve data visualization and reporting skills
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Dataset Overview

Dataset Name

Coffee Shop Sales Dataset

Dataset Type

Transactional Sales Data

Total Records

Approximately 149,000+ transaction records

Main Columns Used

Column Name	Description
transaction_id	Unique transaction identifier
transaction_date	Date of transaction
transaction_time	Time of transaction
transaction_qty	Quantity sold
store_location	Branch/store location
product_category	Product category
product_type	Product type
product_detail	Detailed product description
unit_price	Product unit price
Revenue	Total transaction revenue
Month	Month extracted from transaction date
Hour	Hour extracted from transaction time
Weekday	Day name extracted from date

Data Cleaning & Preparation

Several data preparation steps were completed before analysis:

Data Cleaning Steps

- Checked for missing values
- Removed duplicate records
- Verified transaction IDs
- Standardized date and time formats
- Created calculated revenue column
- Extracted month from transaction date
- Extracted hour from transaction time
- Generated weekday column for trend analysis

Revenue Formula Used

```
=transaction_qty * unit_price
```

Key Performance Indicators (KPIs)

The following KPIs were developed to measure business performance:

1. Total Revenue

Measures total sales generated by the business.

Formula

```
=SUM(Revenue)
```

Business Importance

Shows overall company performance and financial growth.

2. Total Orders

Measures total customer transactions.

Formula

```
=COUNTA(transaction_id)
```

Business Importance

Indicates customer traffic and operational activity.

3. Average Order Value (AOV)

Measures average customer spending per transaction.

Formula

```
=Total Revenue / Total Orders
```

Business Importance

Helps understand customer purchasing behavior.

4. Best Selling Product

Identifies highest-performing products by revenue.

Business Importance

Supports inventory planning and product strategy.

5. Peak Sales Hour

Identifies the busiest hour of the day.

Business Importance

Improves staffing and operational planning.

6. Best Performing Store

Measures highest revenue-generating branch.

Business Importance

Supports branch-level performance evaluation.

Dashboard Analysis

1. Monthly Revenue Analysis

Monthly analysis was performed to identify revenue trends and seasonal patterns.

Key Insights

- Revenue fluctuates across different months
- Some months generate significantly higher sales
- Seasonal customer behavior impacts sales performance

Business Value

Helps management forecast future sales and inventory demand.

2. Product Category Analysis

Sales performance was analyzed by product category.

Categories Included

- Coffee
- Tea
- Bakery
- Chocolate
- Drinking Chocolate

Key Insights

- Coffee products generated the highest revenue
- Beverage categories dominate customer purchases
- Bakery items support additional cross-selling opportunities

Business Value

Supports product optimization and menu strategy.

3. Weekday Revenue Analysis

Sales performance was analyzed by weekday.

Key Insights

- Weekday sales differ significantly
- Some weekdays generate higher customer traffic
- Weekend purchasing patterns vary from weekdays

Business Value

Improves workforce scheduling and promotional planning.

4. Top Products Analysis

Top-selling products were identified using revenue and quantity sold.

Key Insights

- Espresso-based products performed strongly
- Customers prefer premium beverages
- Certain products consistently outperform others

Business Value

Supports inventory management and pricing strategy.

5. Store Performance Analysis

Revenue analysis was conducted across different store locations.

Key Insights

- Some stores significantly outperform others
- Location plays an important role in sales performance
- Customer traffic differs across branches

Business Value

Helps management identify expansion opportunities and operational improvements.

6. Hourly Sales Analysis

Sales were analyzed across different hours of the day.

Key Insights

- Morning hours generate the highest sales
- Sales peak during commuting and breakfast times
- Afternoon sales gradually decline

Business Value

Improves staffing efficiency and operational planning.

Data Visualization Techniques Used

The dashboard includes multiple visualization techniques:

Visualization	Purpose
KPI Cards	Display business metrics
Line Charts	Monthly sales trends
Bar Charts	Product and store comparisons
Pie Charts	Category contribution
Pivot Tables	Data summarization
Slicers	Interactive filtering

Tools & Technologies Used

Tool	Purpose
Microsoft Excel	Data cleaning and dashboard creation
Pivot Tables	Data aggregation
Pivot Charts	Visualization
Excel Formulas	KPI calculations
Data Analysis Techniques	Business insight generation

Business Insights

The project generated several important business insights:

- Coffee products dominate overall revenue generation
 - Morning sales periods are the busiest operational hours
 - Certain branches outperform others consistently
 - Customer purchasing behavior varies by weekday
 - Product category performance can guide future business strategy
 - KPI dashboards improve executive decision-making
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Challenges Faced

Several challenges were encountered during the project:

- Cleaning large transactional datasets
 - Standardizing date and time formats
 - Creating meaningful KPIs
 - Designing user-friendly dashboards
 - Managing large pivot table calculations
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Skills Demonstrated

This project demonstrates several important data analytics skills:

Technical Skills

- Data Cleaning
- Excel Dashboard Development
- Pivot Tables

- Pivot Charts
- KPI Reporting
- Data Visualization
- Business Intelligence
- Sales Analytics

Analytical Skills

- Trend Analysis
 - Business Reporting
 - Problem Solving
 - Insight Generation
 - Performance Analysis
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Future Improvements

The project can be further improved by:

- Integrating Power BI dashboards
 - Adding forecasting models
 - Performing customer segmentation
 - Building predictive analytics models
 - Automating reporting processes
 - Adding profitability analysis
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Conclusion

The Coffee Shop Sales Dashboard project successfully transformed raw transactional data into actionable business insights using Excel-based analytics techniques.

The dashboard provides a comprehensive overview of business performance through KPI tracking, sales trend analysis, store comparison, and product performance evaluation.

This project demonstrates practical business intelligence skills and showcases the ability to perform professional-level sales analytics suitable for real-world business environments.

The analysis supports data-driven decision-making and highlights the importance of dashboard reporting in modern businesses.

References

- Microsoft Excel Documentation
- Business Intelligence Dashboard Practices
- Sales Analytics Reporting Techniques

Author

Prepared as a Data Analytics Portfolio Project.